FOR IMMERSIONS OF ALGEBRAIC SPACES, KOSZUL-REGULAR IMPLIES H1-REGULAR IMPLIES QUASI-REGULAR

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 10549-10557 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 10559-10562, the complete author proof (2 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The schemes-side statement lives in Divisors, not in either pooled chapter — census A=0 here means the label is unique to this file among our two sources, as required.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Divisors, lemma divisors-lemma-regular-quasi-regular-immersion (schemes side); etale-local definitions of the three regularity notions for immersions of spaces.

Source reference. Original lemma 063K: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10562–10562. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $i : Z \to X$ be an immersion of algebraic spaces over S. We have the following implications: i is Koszul-regular $\Rightarrow$ i is H_1-regular $\Rightarrow$ i is quasi-regular.*

Proof. Via the definition this lemma immediately reduces to Divisors, Lemma 063K. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.